Problem Set 4

Instructor solutions

Orders, interpolation, and Fourier inversion

Optional enrichment | 6 problems | 80 points + 10 challenge points | Suggested time: 3–4 hours

Preparation. Course packet Sections 5.10–5.11 and polynomial interpolation from Chapter 4. Use $i^2 = -1$ for complex arithmetic; no calculus or infinite Fourier series is needed.

Instructions. Write on separate paper and use exact arithmetic. You may use poset Möbius inversion and polynomial CRT, except when a problem asks you to establish the relevant formula. Problems 1–3 concern order and counting; Problems 4–6 concern polynomial evaluation and its inverse.

Conventions. For a finite poset P, $\mu_P(x,x) = 1$ and $\mu_P(x,y) = -\sum_{x \le z < y} \mu_P(x,z)$ when $x < y$. For a primitive nth root ω, our Fourier convention is

$$y_j = \sum_{k=0}^{n-1} c_k \omega^{jk}, \qquad c_k = \frac{1}{n} \sum_{j=0}^{n-1} y_j \omega^{-jk}.$$

Over a finite field, $1/n$ means a multiplicative inverse in that field.

Grading note. This is an optional enrichment assignment; subpart scores total 80, with 10 extra challenge points. Sections I and II can also be assigned independently. Accept either recursive subtraction or an explicit Möbius formula where appropriate. Check the direction of the order, the Fourier sign convention, and the factor $1/n$ in addition to arithmetic.

I. Recover exact counts from cumulative counts

1. Four incomparable middle elements [12 points]

Let $P = \{0, a, b, c, d, 1\}$, with $0 < x < 1$ for each $x \in \{a, b, c, d\}$ and no comparisons between distinct middle elements.

(a) Draw the Hasse diagram and list the intervals with exactly two elements. (3)

(b) Compute $\mu_P(0,x), \mu_P(x,1)$ for the middle elements and $\mu_P(0,1)$. (4)

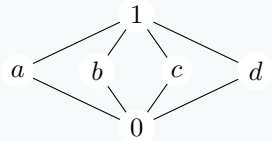
(c) Suppose $G(y) = \sum_{x \le y} F(x)$ and

y	0	a	b	c	d	1
$G(y)$	3	5	8	10	4	25

Recover all six values of F. Check $F(1)$ both by subtraction and by Möbius inversion. (5)

Solution

The two-element intervals are $[0,x]$ and $[x,1]$ for the four middle elements.



Each two-element interval has Möbius value -1, so $\mu_P(0,1) = -(1-4) = 3$. Successive subtraction gives

$$(F(0), F(a), F(b), F(c), F(d), F(1)) = (3, 2, 5, 7, 1, 7).$$

In particular $F(1) = 25 - (3+2+5+7+1) = 7$. Inversion gives the same answer: $F(1) = 3G(0) - G(a) - G(b) - G(c) - G(d) + G(1) = 9 - 27 + 25 = 7$.

2. Membership patterns, including none [14 points]

A universe U has 50 elements, and three subsets satisfy

$$|A| = 24, \quad |B| = 22, \quad |C| = 18, \quad |A \cap B| = 10, \quad |A \cap C| = 8, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$

(a) For $T \subseteq \{A, B, C\}$, let $F(T)$ count elements belonging to exactly the sets in T. Let $G(S)$ count elements in every set in S, with $G(\varnothing) = 50$. Explain why $G(S) = \sum_{T \supseteq S} F(T)$ is an *upper* sum on the subset poset. (4)

(b) Use upper-sum Möbius inversion to recover all eight exact membership counts, including $F(\varnothing)$. State the formula used. (8)

(c) Compute $|A \cup B \cup C|$. Check that your eight counts are nonnegative and sum to 50. (2)

Solution

An element counted by $G(S)$ has an exact membership set T containing S. Since $\mu(S, T) = (-1)^{|T|-|S|}$ on the subset poset, upper inversion gives

$$F(S) = \sum_{T \supseteq S} (-1)^{|T|-|S|} G(T).$$

For example, $F(\{A\}) = 24 - 10 - 8 + 3 = 9$, while $F(\varnothing) = 50 - 24 - 22 - 18 + 10 + 8 + 6 - 3 = 7$. The complete table is

S	$\varnothing$	A	B	C	AB	AC	BC	ABC
$F(S)$	7	9	9	7	7	5	3	3

where AB abbreviates $\{A, B\}$, and similarly for the other labels. All entries are nonnegative and sum to 50. The union has $50 - 7 = 43$ elements, also $24 + 22 + 18 - 10 - 8 - 6 + 3$.

3. Why a partition coefficient can be six [14 points]

Let Π_4 be the partitions of $\{1, 2, 3, 4\}$, ordered by refinement. Write $\hat{0} = 1|2|3|4$ and $\hat{1} = 1234$. Partitions have block-size types $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, or 4.

(a) Count the partitions of each type. (4)

(b) Compute $\mu_{\Pi_4}(\hat{0}, \pi)$ for each type, ending with $\mu_{\Pi_4}(\hat{0}, \hat{1})$ by the recurrence. *Hint:* for type $2+2$, the interval is a product of two two-element chains; for type $3+1$, it is Π_3. (6)

(c) There are a colors. Let $F(\rho)$ count colorings of four labeled positions whose equality partition is exactly ρ, and let $G(\pi)$ count colorings constant on every block of π. Explain why $G(\pi) = a^{|\pi|} = \sum_{\rho \geq \pi} F(\rho)$, where $|\pi|$ counts blocks. Invert to count colorings with all four colors distinct. Give the result for $a = 5$ and compare with direct counting. (4)

Solution

The counts are obtained by choosing a pair, choosing an unordered pairing, or choosing a triple, as applicable:

type	$1+1+1+1$	$2+1+1$	$2+2$	$3+1$	4
number	1	6	3	4	1
$\mu(\hat{0}, \pi)$	1	-1	1	2	-6

A two-element chain contributes -1, so a product of two contributes 1. For Π_3, the top coefficient is $-(1-3) = 2$. Finally the proper elements below $\hat{1}$ contribute $1 - 6 + 3 + 8 = 6$, so its coefficient is -6.
A coloring constant on blocks of π may give the same color to different blocks; its equality partition is therefore a coarsening $\rho \geq \pi$. Upper inversion gives

$$F(\hat{0}) = \sum_{\pi \in \Pi_4} \mu(\hat{0}, \pi) a^{|\pi|} = a^4 - 6a^3 + 11a^2 - 6a = a(a-1)(a-2)(a-3).$$

For $a = 5$ this is 120. Directly, the four labeled positions have $a, a-1, a-2, a-3$ successive color choices.

II. Recover coefficients from values

4. Lagrange selectors become Fourier inversion [12 points]

Let $n \geq 2$ and let $\omega \in \mathbb{C}$ have order exactly n.

(a) Use a finite geometric sum to show, for every integer m,

$$\sum_{j=0}^{n-1} \omega^{jm} = \begin{cases} n, & n \mid m, \\ 0, & n \nmid m. \end{cases}$$

Explain where primitivity of ω enters. (4)

(b) Define $\ell_j(X) = \frac{1}{n}\sum_{k=0}^{n-1} \omega^{-jk} X^k$. Show that $\ell_j(\omega^r) = 1$ for $r = j$ and 0 otherwise, for $0 \leq r, j < n$. Explain why this identifies the Lagrange basis. (4)

(c) If $f(X) = \sum_{k=0}^{n-1} c_k X^k$ and $y_j = f(\omega^j)$, expand $f = \sum_j y_j \ell_j$ to derive the inverse Fourier formula. Explain the role of the degree restriction. (4)

Solution

Primitivity says $\omega^m = 1$ exactly when $n \mid m$. In that case all n terms equal 1; otherwise their sum is $(1 - (\omega^m)^n)/(1 - \omega^m) = 0$. Thus $\ell_j(\omega^r) = n^{-1}\sum_k \omega^{k(r-j)}$ has the stated values. These polynomials have degree less than n, so uniqueness of interpolation at n distinct points makes them the Lagrange selectors. Expanding the interpolant gives

$$f(X) = \sum_{j=0}^{n-1} y_j \left(\frac{1}{n} \sum_{k=0}^{n-1} \omega^{-jk} X^k \right) = \sum_{k=0}^{n-1} \left(\frac{1}{n} \sum_{j=0}^{n-1} y_j \omega^{-jk} \right) X^k.$$

Comparing coefficients proves the inverse. Without the degree restriction, values determine only the remainder modulo $X^n - 1$; for example, f and $f + (X^n - 1)$ have identical values at all these roots.

5. The same four-point transform in two fields [16 points]

Take $f(X) = 2 - X + 3X^3$, with coefficient vector $(2, -1, 0, 3)$. For each field below, compute the four evaluations and then recover the coefficients using the inverse transform, showing each inverse sum.

(a) Over $\mathbb{C}$, use $\omega = i$ and list values in the order $1, i, -1, -i$. (8)

(b) Over $\mathbb{F}_{13}$, prove that $\omega = 5$ has order 4. List values in the order $1, 5, 12, 8$. Find both ω^{-1} and 4^{-1} before applying the inverse formula. Interpret the recovered coefficient of X. (8)

Solution

Over $\mathbb{C}$, the value vector is $(4, 2 - 4i, 0, 2 + 4i)$. The four inverse sums give

$$\begin{aligned} c_0 &= \tfrac{1}{4}[4 + (2 - 4i) + 0 + (2 + 4i)] = 2, \\ c_1 &= \tfrac{1}{4}[4 - i(2 - 4i) - 0 + i(2 + 4i)] = -1, \\ c_2 &= \tfrac{1}{4}[4 - (2 - 4i) + 0 - (2 + 4i)] = 0, \\ c_3 &= \tfrac{1}{4}[4 + i(2 - 4i) - 0 - i(2 + 4i)] = 3. \end{aligned}$$

Solution (continued)

In $\mathbb{F}_{13}$, $5^2 = 12 = -1$ and $5^4 = 1$, so the order is 4. Its inverse is 8, and $4^{-1} = 10$. The evaluations are $(4, 8, 0, 9)$; for instance $f(5) = 2 - 5 + 3 \cdot 8 = 21 \equiv 8$. The inverse calculation is

$$\begin{aligned} c_0 &= 10(4 + 8 + 0 + 9) = 210 \equiv 2, \\ c_1 &= 10(4 + 8 \cdot 8 + 12 \cdot 0 + 5 \cdot 9) = 1130 \equiv 12, \\ c_2 &= 10(4 + 12 \cdot 8 + 0 + 12 \cdot 9) = 2080 \equiv 0, \\ c_3 &= 10(4 + 5 \cdot 8 + 12 \cdot 0 + 8 \cdot 9) = 1160 \equiv 3. \end{aligned}$$

All congruences are modulo 13. The recovered vector is $(2, 12, 0, 3)$, which is $(2, -1, 0, 3)$ in this field.

6. Why Fourier multiplication wraps around [12 points]

Work over $\mathbb{C}$ with four evaluation points $1, i, -1, -i$. Let $f(X) = 1 + 2X + X^3$ and $g(X) = 2 + X^2$.

(a) Multiply ordinarily, then reduce modulo $X^4 - 1$ to find the unique remainder h of degree less than 4. (4)

(b) Compute the two value vectors, multiply them entry by entry, and invert to recover the coefficient vector of h. (4)

(c) Write the cyclic convolution formula for the coefficient of X^r in h, using coefficient vectors $c = (1, 2, 0, 1)$ and $d = (2, 0, 1, 0)$. Explain why the X^5 term contributes to X, and how polynomial CRT justifies multiplication of values. (4)

Solution

The ordinary product and its remainder are

$$fg = 2 + 4X + X^2 + 4X^3 + X^5, \qquad h = 2 + 5X + X^2 + 4X^3.$$

The value vectors are $(4, 1 + i, -2, 1 - i)$ and $(3, 1, 3, 1)$; their entrywise product is $y = (12, 1 + i, -6, 1 - i)$. The four inverse numerators, before division by 4, are

$$\begin{aligned} 12 + (1 + i) - 6 + (1 - i) &= 8, \\ 12 - i(1 + i) + 6 + i(1 - i) &= 20, \\ 12 - (1 + i) - 6 - (1 - i) &= 4, \\ 12 + i(1 + i) + 6 - i(1 - i) &= 16. \end{aligned}$$

Hence the coefficient vector is $(2, 5, 1, 4)$, as above. In general it is $(c * d)_r = \sum_{k=0}^{3} c_k d_{(r-k) \bmod 4}$. Since $X^4 = 1$ in the quotient, $X^5 = X$, so its coefficient wraps into the coefficient of X. The four linear factors of $X^4 - 1$ are pairwise coprime. Polynomial CRT identifies the quotient ring with four copies of $\mathbb{C}$ via evaluation, and multiplication there is entry by entry.

Optional challenge: recover the ordinary product by adding zeros [10 extra points]

Use the same f and g as in Problem 6.

(a) What is the smallest transform length N for which the remainder of fg modulo $X^N - 1$ must equal the ordinary product? Explain why length 8 also works. (4)

(b) Work now over $\mathbb{F}_{17}$. Show that 2 has order 8 and find the inverse normalization factor 8^{-1}. (3)

(c) Append zeros to make both coefficient vectors length 8. Explain why evaluating at the powers of 2, multiplying values, and inverting recovers the ordinary product over $\mathbb{F}_{17}$. Give the resulting length-8 coefficient vector; you need not compute the eight-point transforms explicitly. (3)

Solution

The product has degree 5, so $N = 6$ is the smallest possible length; any $N > 5$, including 8, prevents terms from wrapping. In $\mathbb{F}_{17}$, $2^4 = 16 = -1$ and $2^8 = 1$, so 2 has order 8. The normalization is $8^{-1} = 15$, since $8 \cdot 15 = 120 \equiv 1 \pmod{17}$. The padded vectors are $(1, 2, 0, 1, 0, 0, 0, 0)$ and $(2, 0, 1, 0, 0, 0, 0, 0)$. CRT and Fourier inversion recover their product modulo $X^8 - 1$. Its degree is only 5, so it is already the ordinary product, with vector $(2, 4, 1, 4, 0, 1, 0, 0)$ in $\mathbb{F}_{17}^8$.